

Paulina Hoyos

Curriculum Vitae

The University of Texas at Austin
Department of Mathematics
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Education

Ph.D. in Mathematics

The University of Texas at Austin
Dissertation:
Advisor: Joe Kileel

2021–Spring 2027 (Expected)

M.S. in Mathematics

Universidad Nacional de Colombia, Medellín, Colombia

2017–2021

B.S. in Physics

Universidad de los Andes, Bogotá, Colombia

2012–2016

Research Experience

Senior Research Scientist Intern

IBM Research, Thomas J. Watson Research Center, Yorktown Heights, NY
Mentors: Shashanka Ubaru and Lior Horesh

Summer 2025

Theory and Mathematical Sciences Intern

IBM Research, Thomas J. Watson Research Center, Yorktown Heights, NY
Mentors: Shashanka Ubaru and Lior Horesh

Summer 2024

Research Interests

Applied and computational mathematics: manifold learning and dimensionality reduction in the presence of symmetries; group actions and representation theory in data analysis; randomized numerical linear algebra and tensor methods; spectral graph theory and discrete heat kernels; applications to cryo-EM and structural biology.

Publications

Authors marked * contributed equally.

Entries marked † follow the alphabetical-author convention standard in mathematics.

Peer-reviewed papers

1. M. Fraiman*, **P. Hoyos***, T. Bendory, J. Kileel, O. Mickelin, N. Sharon, and A. Singer. *SO(3)-invariant PCA with application to molecular data*. IEEE International Symposium on Biomedical Imaging (ISBI), 2026. arXiv: 2510.18827.
2. †A. Castillo, J. Haddock, I. Hartsock, **P. Hoyos**, L. Kassab, A. Kryshchenko, K. Larripa, D. Needell, S. Suryanarayanan, and K. Yacoubou-Djima. *Block Gauss-Seidel methods for t-product tensor regression*. Numerical Algorithms **103** (2026), 557–589. (arXiv: 2503.19155).

3. [†]A. Castillo, J. Haddock, I. Hartsock, **P. Hoyos**, L. Kassab, A. Kryshchenko, K. Larripa, D. Needell, S. Suryanarayanan, and K. Yacoubou-Djima. *Randomized Kaczmarz methods for t-product tensor linear systems with factorized operators*. BIT Numerical Mathematics **65** (2025), article 38. arXiv: 2412.10583.
4. [†]A. Castillo, J. Haddock, I. Hartsock, **P. Hoyos**, L. Kassab, A. Kryshchenko, K. Larripa, D. Needell, S. Suryanarayanan, and K. Yacoubou-Djima. *Randomized Iterative Methods for Tensor Regression Under the t-Product*. In *Advances in Data Science*, Association for Women in Mathematics Series, vol. 37, Springer, 2025, pp. 3–38.
5. E. Rosen, **P. Hoyos**, X. Cheng, J. Kileel, and Y. Shkolnisky. *The G-invariant graph Laplacian Part I: Convergence rate and eigendecomposition*. Applied and Computational Harmonic Analysis **71** (2024), 101637. arXiv: 2303.17001.
6. [†]C. Cadavid, **P. Hoyos**, J. Jorgenson, L. Smajlović, and J. Vélez. *Discrete diffusion-type equation on regular graphs and its applications*. Journal of Difference Equations and Applications, 29:4, 455-488, 2023.
7. [†]C. Cadavid, **P. Hoyos**, J. Jorgenson, L. Smajlović, and J. Vélez. *On an approach for evaluating certain trigonometric character sums using the discrete time heat kernel*. European Journal of Combinatorics, Volume 108, 2023.
8. N. Morales-Durán, A.F. Vargas, **P. Hoyos-Restrepo**, and P. Bargueño. *Simple regular black hole with logarithmic entropy correction*. Eur. Phys. J. C, **76**: 559, 2016.

Preprints

1. Y. Vigder*, **P. Hoyos***, D. Thong, J. Andén, J. Kileel, and A. Moscovich. *Group Invariant Spectral Embedding*. arXiv: 2607.08987.
2. **P. Hoyos**, S. Ubaru, D. Huh, V. Kalantzis, K. L. Clarkson, M. E. Kilmer, H. Avron, and L. Horesh. *Exact Symmetry as Algebra: A Machine-Verified Tensor Calculus that Enforces Physical Selection Rules*. arXiv: 2605.20440.
3. [†]C. Cadavid, **P. Hoyos**, J. Jorgenson, L. Smajlović, and J. Vélez. *Diffusion Computation versus Quantum Computation: A Comparative Model for Order Finding and Factoring*. arXiv: 2601.02518.
4. [†]A. Castillo, J. Haddock, I. Hartsock, **P. Hoyos**, L. Kassab, A. Kryshchenko, K. Larripa, D. Needell, S. Suryanarayanan, and K. Yacoubou Djima. *Quantile-Based Randomized Kaczmarz for Corrupted Tensor Linear Systems*. arXiv: 2503.18190.

Presentations

Invited Talks

1. *Group-Algebraic Tensors: Optimal Decompositions for Equivariant Learning*. SIAM Conference on Mathematics of Data Science, Minisymposium on Geometric and Algebraic Foundations of Deep Learning, Salt Lake City, UT, November 2026.
2. *Synthetic Molecular Data Generation via SO(3)-Invariant PCA*. SIAM TX-LA Sectional Meeting, Minisymposium on Applications of Algebraic Geometry, Southern Methodist University, Dallas, TX, October 2026.
3. *Group-Algebraic Tensors: Optimal Decompositions for Equivariant Learning*. AMS 2026 Fall Southeastern Sectional Meeting, Special Session on Applied Algebra in Statistics, Optimization, and Machine Learning, Kennesaw, GA, October 2026.
4. *Synthetic Molecular Data Generation via SO(3)-Invariant PCA*. SIAM 2026 Annual Meeting, Minisymposium on Algebraic Geometry and Computer Vision, Cleveland, OH, July 2026.
5. *Group-Algebraic Tensors and Equivariant Compression of Data*. SIAM 2026 Annual Meeting, Minisymposium on Tensors for All: Algebraic, Geometric, and Computational Perspectives of Multilinear Algebra, Cleveland, OH, July 2026.
6. *Manifold Learning in the Presence of Symmetries*. Codes and Expansions (CodEx) Seminar, virtual, August 2025.
7. *An Extension of the Tensor Randomized Kaczmarz Algorithm to Problems with Factorized Operator or Data*. SIAM Conference on Mathematics of Data Science, Minisymposium on Randomized Iterative Algorithms for Large-Scale Matrix and Tensor Data, Atlanta, GA, October 2024.

8. *Manifold Learning in the Presence of Symmetries*. Geometry Seminar, Texas A&M University, College Station, TX, March 2024.

Contributed Talks and Posters

1. *Manifold Learning in the Presence of Symmetries: the G-Invariant Graph Laplacian*. SIAM Conference on Mathematics of Data Science, Poster Session, Atlanta, GA, October 2024.
2. *Randomized Algorithms for Tensor Problems with Factorized Operator or Data*. Connecting Higher-Order Statistics and Symmetric Tensors Workshop, ICERM, Brown University, Providence, RI, January 2024.
3. *Trigonometric Identities from the Heat Kernel on Graphs*. Texas Women in Mathematics Symposium (TWIMS), The University of Texas at Austin, Austin, TX, March 2023.
4. *The Discrete-Time Heat Equation on Graphs as a Computational Tool*. Data and Algebra Seminar, Department of Mathematics, The University of Texas at Austin, Austin, TX, February 2022.
5. *The Heat Equation and Minimal Morse Functions*. Uniandes SRI on Geometry and Theoretical Physics, Villa de Leyva, Colombia, July 2019.
6. *The Heat Equation and Minimal Morse Functions*. Segundo Encuentro de Álgebra y Topología, Universidad Nacional de Colombia, Medellín, Colombia, May 2019.
7. *The Ayón-Beato–García Regular Black Hole and Topology Change*. Latin-American Workshop on High Energy Physics: Particles and Strings II, Havana, Cuba, July 2016.
8. *The Ayón-Beato–García Regular Black Hole and Topology Change*. Eighth Mathematical Physics School: Random Geometries, Poster Session, Universidad de los Andes, Bogotá, Colombia, May 2016.
9. *The GNS Construction*. Workshop on the Mathematical Structure and Foundations of Quantum Physics, Universidad de los Andes, Bogotá, Colombia, June 2015.

Participation in Academic Programs

1. SQuaRE Program on Randomized Gauss-Seidel and Column-Slice-Action Methods for Tensor Problems, American Institute of Mathematics (AIM), Pasadena, CA, 2024–present.
2. Collaborate@ICERM Program, ICERM, Brown University, Providence, RI, August 2024.
3. Collaborative Research Workshop for Women in Data Science and Mathematics (WISDM), IPAM, UCLA, Los Angeles, CA, August 2023.
4. Mathematics of Big Data: Sketching and (Multi-)Linear Algebra, SLMATH Summer Graduate School, IBM Almaden, San Jose, CA, July 2023.
5. IMPA Summer Program, Rio de Janeiro, Brazil, January–March 2018.

Honors and Awards

1. University Graduate Continuing Fellowship, Graduate School, The University of Texas at Austin, Fall 2024–Spring 2025.
2. Graduate School Summer 2023 Fellowship, The University of Texas at Austin, Summer 2023.
3. Laureate distinction for M.Sc. thesis *On the Discrete Heat Equation and Kolmogorov Complexity Theory*, Universidad Nacional de Colombia, Medellín, Colombia, February 2021.
4. Academic Excellence Fellowship, Faculty of Sciences, Universidad Nacional de Colombia, Medellín, Colombia, April 2019.
5. Summer Program Scholarship, IMPA, Rio de Janeiro, Brazil, January 2018.
6. *Bachilleres por Colombia* “Mario Galán Gómez” scholarship for undergraduate studies, Ecopetrol S.A., Colombia, August 2011.

Teaching

Teaching Assistant, The University of Texas at Austin, Austin, TX

1. M427L Advanced Calculus for Applications II (upper division), Summer 2026.
2. M362K Probability I (upper division), Fall 2021, Spring 2022, and Spring 2024.
3. M383C Methods of Applied Mathematics (graduate), Fall 2023.
4. M358K Applied Statistics (upper division), Spring 2023.
5. M427J Differential Equations with Linear Algebra (upper division), Fall 2022.

Other Teaching, The University of Texas at Austin, Austin, TX

1. *Fourier Analysis on Finite Groups*, graduate student summer minicourse, Department of Mathematics, July 2022.

Instructor, Universidad Nacional de Colombia, Medellín, Colombia

1. Differential Calculus, Fall 2020, Spring 2019, and Fall 2018.
2. Basic Mathematics (Precalculus), Spring 2018.

Teaching Assistant, Universidad Nacional de Colombia, Medellín, Colombia

1. Differential Calculus, July–December 2017.

Teaching Assistant, Universidad de los Andes, Bogotá, Colombia

1. Prephysics, Spring 2016.
2. *Clínica de Problemas de Física*, Fall 2014.

Mentoring

1. Directed Reading Program (DRP), Department of Mathematics, The University of Texas at Austin, Fall 2023. Topic: Randomized Algorithms in Numerical Linear Algebra.
2. Directed Reading Program (DRP), Department of Mathematics, The University of Texas at Austin, Spring 2023. Topic: Introduction to Graph Theory.

Outreach

1. Organizer, Sunday Morning Math Group (SMMG), The University of Texas at Austin, Fall 2025–Spring 2026 and Fall 2026–Spring 2027.
2. Volunteer, Sonia Kovalevsky Day and Math Fun Day with Girl Scouts of Central Texas, The University of Texas at Austin, November 2023.
3. *Secure Communication with Cryptography*. Sunday Morning Math Group (SMMG), The University of Texas at Austin, October 2023.
4. Volunteer, Department of Mathematics Stand, Girl Day at UT Austin, The University of Texas at Austin, February 2023.
5. *Introduction to Quantum Computing: What Is and What Isn't a Quantum Computer?* Sunday Morning Math Group (SMMG), The University of Texas at Austin, January 2023.
6. *Random Walks on Graphs*. Sunday Morning Math Group (SMMG), The University of Texas at Austin, November 2022.
7. *Random Walks on Graphs*. Research in Math Club, LASA High School, Austin, TX, October 2022.

Leadership

1. Organizer, Junior Applied Mathematics Seminar, Department of Mathematics, The University of Texas at Austin, Spring 2024.
2. Member, Graduate Representative Committee (GRC), Department of Mathematics, The University of Texas at Austin, Fall 2023–Spring 2024.

Professional Memberships

American Mathematical Society (AMS)

Society for Industrial and Applied Mathematics (SIAM)